4. Proposed Methodology

In this paper we propose an end-to-end system that utilizes state-of-the-art deep learning architectures, specifically CNNs and BiLSTMs, for sequence prediction on check images from the CENPARMI dataset. The system incorporates a comprehensive verification pipeline, starting with the extraction of courtesy and legal amount regions, followed by the recognition of courtesy and legal amounts using a CNN-BiLSTM model, then conversion of the legal amount into its numerical representation using a natural language processing (NLP) algorithm, and finally, a verification step through cross-comparison.

The proposed methodology progresses through several phases to achieve the most accurate and reliable verification process. These steps are logically structured, with each stage building upon the previous one to establish a robust, systematic framework for the automatic verification of courtesy and legal amounts. The methodology comprises the following phases:

- Dataset preperation: Preparing a dataset for automated extraction, which includes annotating checks to align with the specific requirements of the designated task.
- Courtesy and legal amount extraction: Applying deep learning models for automatic extraction of courtesy and legal amounts' fields on Arabic checks
- Courtesy amount recognition: Automatic recognition of the extracted courtesy amount which represents the issued amount written in numerical representation.
- Legal amount recognition: Automatic recognition of the extracted legal amount which represents the issued amount in written format.
- Legal amount conversion: convert the extracted legal amount to numerical representation for the purpose of comparing it with the extracted courtesy amount.
- Improve courtesy amount recognition using converted legal amounts, and vice versa, by exploiting their complementary error patterns. Courtesy amount errors mainly involve insertion, deletion, or omission of digits (e.g., "1,000" → "1,000,000"), though digit-level recognition remains reliable with rare substitutions. Conversely, legal amount recognition in Arabic is more prone to word-level substitution errors (e.g.,

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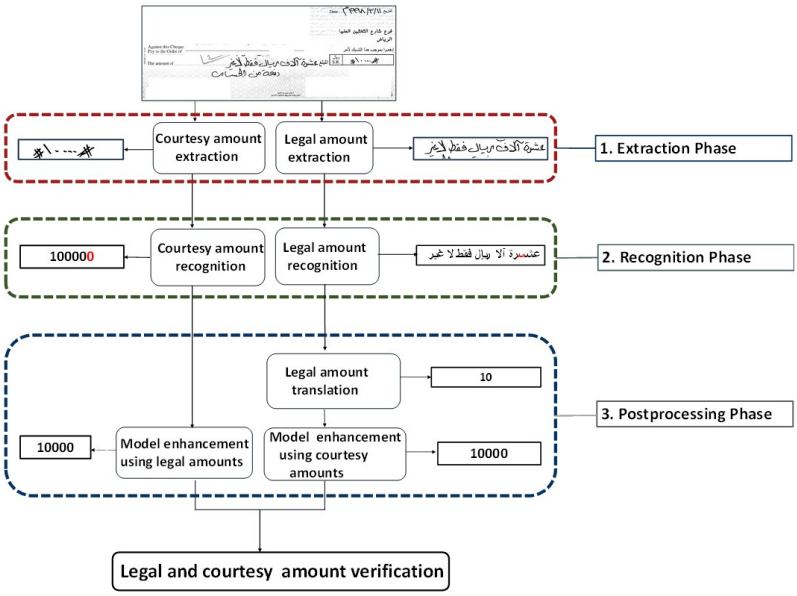

Figure 5: The methodology steps for extracting and verifying courtesy and legal amounts.

ستة vs. سبعة), while words like مئة, ألف, and مليون serve as strong cues for numerical magnitude. By cross-validating predicted legal and courtesy amounts, the system can detect and correct inconsistencies, enhancing the overall reliability of check amount recognition.

- Verify the legal and courtesy amounts by comparing the courtesy amount with the corresponding converted legal amount.

These steps are organized into three overarching phases: the extraction phase, the recognition phase, and the postprocessing phase, which together lead to the final verification of the check amounts. Fig. 5 illustrates the complete workflow of the proposed methodology. Each phase will be discussed in detail in the subsequent sections, including implementation strategies and an in-depth analysis of the results.

5. Courtesy and Legal Amounts Extraction

The first step in this work is to extract courtesy and legal amounts from Arabic checks. Despite the availability of pre-extracted legal and courtesy

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amounts in the dataset, we aimed to provide an end-to-end pipeline that begins with courtesy and legal fields extraction, followed by processing, recognition, and finally, verification of the legal and courtesy amounts. However, the check images in the CENPARMI dataset are not labeled for the extraction task. In this section, we will thoroughly explain the labeling procedure for the courtesy and legal amounts, followed by the proposed model, and finally, present a detailed analysis of the results.

5.1. Dataset and Check Images Labeling

In this work, we used the Arabic checks dataset developed by the Center for Pattern Recognition and Machine Intelligence (CENPARMI) Al-Ohali et al. (2003). The dataset contains check images sourced from Alrajhi Bank in Riyadh during the period 1998–2003, ensuring that the data reflect realistic scenarios encountered in daily financial transactions. Moreover, it captures a wide diversity in handwriting styles and formats, including variations such as the use of delimiters or commas in the courtesy amount field, as well as the presence of noise like pen stains or stamps on the check images.

The dataset comprises 2,498 real Arabic bank check images, of which 1,779 are used for training and the remaining 719 for testing. It is important to note that the check images in this dataset are not directly labeled for the legal and courtesy amount extraction task.

To address this, we manually annotated the training set by labeling the bounding boxes surrounding the courtesy and legal amount fields in the 1,779 training images.

5.2. Extraction Model

Due to the limited availability of samples for training a deep learning model from scratch to extract the courtesy and legal fields from Arabic check images, we fine-tuned the pretrained YOLOv8 model by Varghese and M. (2024), which is a state-of-the-art deep learning model designed for object detection tasks. YOLO v8 employs CNNs to capture image features at multiple levels, which are then concatenated to generate bounding boxes around potential objects within the image. The model consists of three main blocks. The first is the backbone, responsible for extracting features and providing a hierarchical representation of the input image. The second block is the neck, which collects the feature maps generated by the backbone and concatenates them. The third block is the head, which produces the final output and

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generates bounding boxes around the desired objects. The YOLO model has been trained on a variety of datasets to recognize a wide range of objects.

The labeled training set was divided into two partitions: one for system training, comprising 1,512 images (85%), and another for model evaluation, consisting of 267 images (15%). The YOLOv8 model was trained for 100 epochs with a batch size of 16 using AdamW optimization (learning rate: 0.001, momentum: 0.9) and L2 regularization (weight decay: 0.0005). Training employed automatic mixed precision (AMP) on a Tesla T4 GPU, with input images resized to 640×640 pixels.

After fine-tuning the YOLO model for the extraction of courtesy and legal amounts using the training data, we applied the optimized model to the testing data to automatically extract the fields of courtesy and legal amounts.

5.3. Results and Analysis

5.3.1. Performance Metrics

The model's performance was evaluated using accuracy at Intersection over Union (IoU) thresholds of 50%, 75%, and 90%. Accuracy was determined by counting the number of samples for which the IoU between the predicted and true values exceeded or equaled the specified threshold. This count was then divided by the total number of samples (267) to calculate the accuracy. The formula for this calculation is presented in Equation 1.

$$\text{Accuracy}_t = \frac{\text{Samples with IoU} \geq t}{\text{Total number of samples}} \tag{1}$$

Where t represents the predfined three thresholds 50%, 75% and 90%.

Furthermore, to comprehensively assess the overall extraction performance, we calculated the mean Intersection over Union (IoU) across all samples, as defined by Equation 2:

$$\text{Mean IoU} = \frac{1}{N} \sum_{i=1}^{N} \text{IoU}_i \tag{2}$$

5.3.2. Extraction Results and Analysis

Table 1 presents the results of the fine-tuned YOLO model evaluated on 267 images. While 67.79% of the courtesy fields have an IoU of 90% or higher between the predicted and ground truth bounding boxes, only 25.84%

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Table 1: Results of courtesy and legal amounts extraction from check images, presenting the performance metrics including accuracy at IoU thresholds of 50% and 75%, 90% .

	Accuracy (IoU–50)	Accuracy (IoU–75)	Accuracy (IoU–90)	Mean IoU
courtesy extraction	100.00%	99.25%	67.79%	91.00%
legal extraction	99.63%	88.76%	25.84%	83.70%

of the legal fields meet this threshold. This sharp drop indicates that the legal amount region is more challenging to localize precisely. The decrease in accuracy at higher IoU thresholds is expected, as even slight misalignments can cause predictions to be counted as incorrect. Fig. 6 illustrates examples of legal field predictions where the intersection between the predicted and true bounding boxes is less than 90%. These cases demonstrate how minor shifts in bounding box position can significantly impact evaluation results, particularly for legal amounts.

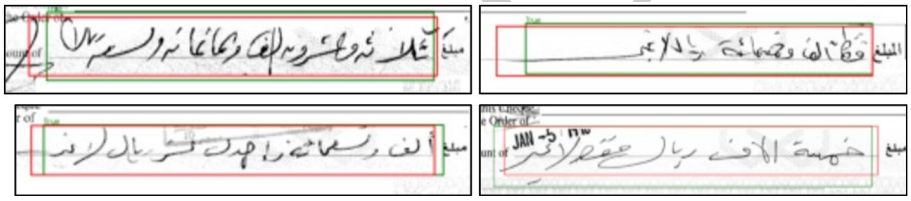

Figure 6: Legal amount extracted and considered as false prediction under IOU 90%. The predicted bounding box shown in red and the actual bounding box in green.

To further analyze the model's performance, we identified four samples with the lowest Intersection over Union (IoU) for legal field extraction. Fig. 7 presents these samples, including both the true and predicted bounding boxes. The figure illustrates that while the entire legal amounts were successfully extracted, the labeling process either failed to fully cover the target area or, in some cases, overcovered it, resulting in low IoU values. In other instances, the legal amounts included non-value words (e.g., the Arabic word 'شكرا' ['thank you']), which were intentionally excluded from annotations but were still captured by the predicted bounding boxes. Despite these discrepancies, the extractions remained sufficiently accurate for recognition tasks, as all target amounts were fully covered.

After fine-tuning the YOLO model and analyzing its performance, we

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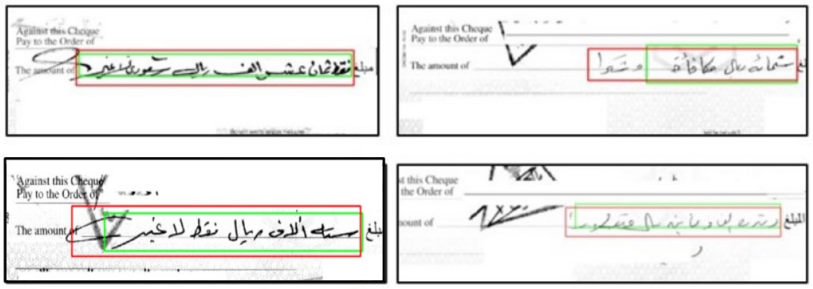

Figure 7: Four legal fields' samples with the lowest IoU values, showing the predicted bounding boxes in red and the true bounding boxes in green.

applied the extraction process for courtesy and legal amounts on the testing dataset, which consists of 719 check images.

5.3.3. Comparison with Previous Work

Table 2 summarizes previous studies on check field extraction using datasets from different countries, including Arabic, Indian, and Brazilian checks. Most of these studies relied on handcrafted or rule-based methods, achieving promising results through techniques such as projection profiles, Hough Transforms, or template matching. However, such approaches often require manual feature engineering and are sensitive to layout variations. In contrast, our deep learning–based model offers greater flexibility and generalization, effectively detecting fields without the need for predefined structural rules or handcrafted features.

6. Courtesy Amount Recognition

This section presents the model developed for recognizing courtesy amounts written in Indian digits, as typically found in Arabic bank checks. The input to this model consists of courtesy amount fields extracted in the previous detection step.

The recognition pipeline utilizes a CNN-BiLSTM-CTC architecture, illustrated in Fig. 8. The CNN layers extract spatial features, followed by BiLSTM layers that capture the sequential dependencies of the digit sequences in both directions. The Connectionist Temporal Classification (CTC) loss enables alignment between the predicted and ground-truth sequences without

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Table 2: Comparison of previous field extraction methods with our work.

Authors	**Year**	**Language / Region**	**Methodology**	**Results / Accuracy**
Ahmad and Mahmoud (2012)	2012	Arabic (CENPARMI)	Rule-based projection + handcrafted evaluation on full dataset	99.5% (courtesy), 99.4% (legal)
Samoud et al. (2009)	2012	Arabic (CENPARMI)	Hough Transform + Mathematical Morphology (dilation)	98.2% (courtesy), 91.8% (legal)
Vishnuvardhan et al. (2021)	2021	Indian (multi-format)	CNN + YOLOv3 object detection	mAP 86.6% (multiple fields)
Palacios Hielscher et al. (2016)	2002	Brazilian	Fixed-layout ROI detection using template matching	High accuracy (qualitative)
Ours (YOLOv8)	**2025**	**Arabic (CENPARMI)**	**YOLOv8 object detection model**	**100% (courtesy), 99.6% (legal)**

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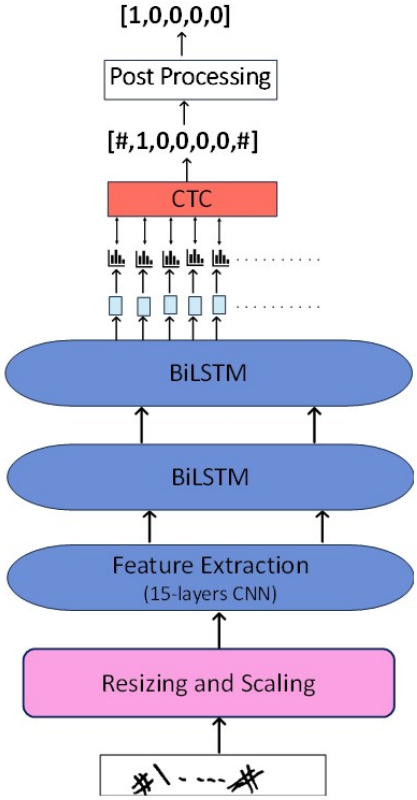

Figure 8: Proposed model for courtesy amount recognition using CNN, BiLSTM and CTC for sequence prediction.

requiring explicit segmentation. Input images undergo minimal preprocessing—resizing and normalization—while deliberately preserving subtle digit features like the dot-shaped zero.

The automatically extracted courtesy regions were divided into 85% for training (1,512 images) and 15% for validation (approximately 267 images). The model was then trained for 500 epochs, with early stopping implemented using a patience of 20 epochs. RMSprop was employed as the optimization function with a learning rate of 1e-4.

6.1. Results and Comparison with Previous Work

To evaluate the effectiveness of our proposed model, we conducted experiments using a test set comprising 719 samples. The model, based on a

CNN-BiLSTM architecture trained with CTC loss, was evaluated at both the digit and amount levels to provide a comprehensive performance assessment.

At the digit level, performance was measured using the Levenshtein distance, which calculates the minimum number of insertions, deletions, and substitutions required to transform the predicted sequence into the ground truth. The digit-level accuracy was computed as shown in Equation 3. Using this metric, the model achieved a digit-level accuracy of 96.1%.

$$Accuracy = (1 - I + D + S/N) * 100 \tag{3}$$

At the amount level, a prediction was considered correct only if the entire predicted sequence exactly matched the ground truth. Based on this strict criterion, the model correctly predicted 613 out of 719 amounts, resulting in an amount-level accuracy of 85

Table 3 presents a detailed analysis of the model's predictions, categorized by the number of digit-level errors.

Table 3: Results of Courtesy Amount Recognition, showing the number of predictions without errors, with one error, with two errors, and with more than two errors, along with the overall accuracy.

Number of samples without errors	613(85%)
Number of samples with one error	90(12%)
Number of samples with two errors	12(2%)
Number of samples with more than 2 errors	4(1%)
Accuracy (digit level)	96.1%

Fig. 9 illustrates the confusion matrix for the CNN-BiLSTM-CTC model, providing a detailed analysis of its performance in recognizing individual digits. The results reveal that the majority of errors consist of substitution and deletion errors, with digit *0* being the most affected (29 deletions, 5 substitutions), followed by digit *1*. This pattern is particularly notable given *0*'s visual similarity to delimiters separating numbers. Despite this challenge, the model achieves robust accuracy for *0*, with 1,577 correct predictions.

This error pattern—particularly the frequent of substitution/deletion errors for digits *0* and *1*—is critical for our postprocessing discussion, where we used recognized legal amounts to correct the predicted courtesy amount.

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Notably, the model maintains strong performance across all digit classes, including infrequent digits like *7* and *9*, which exhibit near-zero deletion errors and minimal substitutions. This demonstrates effective generalization regardless of sample frequency in the training data.

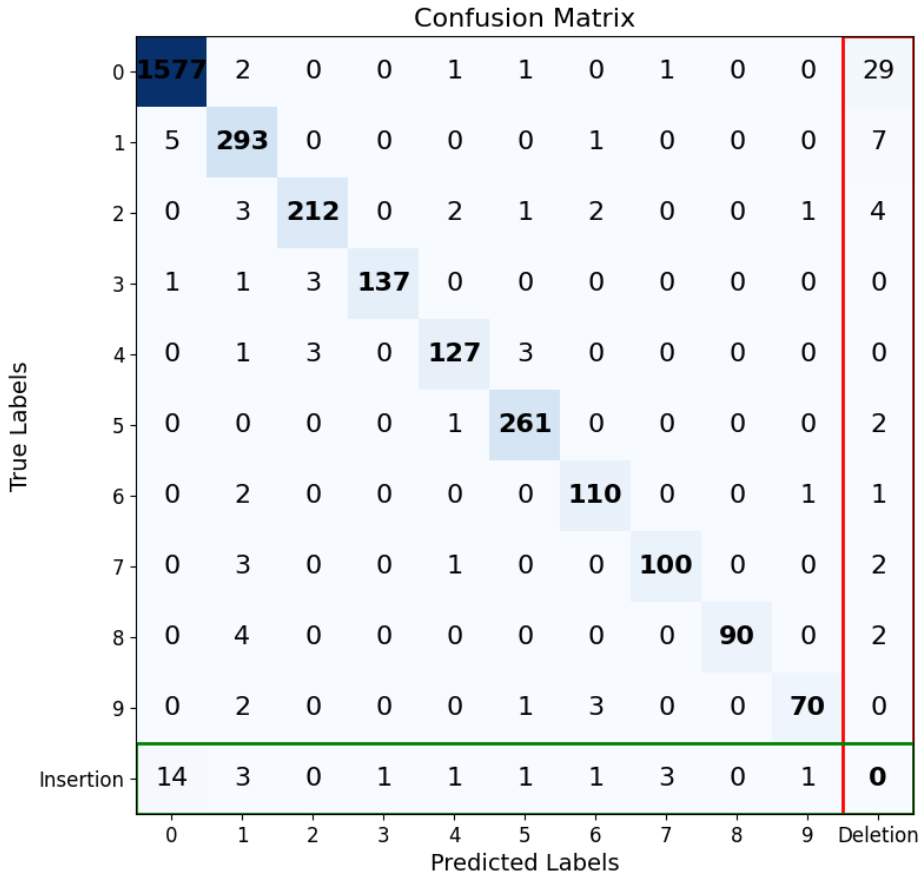

Figure 9: Confusion matrix for the CNN-BiLSTM-CTC model evaluated on the testing dataset, illustrating the model's performance in classifying each digit, including the insertions and deletions for each digit.

Furthermore, we compared our model with the hybrid rule-based and machine learning method proposed by Ahmad Ahmad (2023), evaluated on a shared subset of 625 samples. As shown in Table 4, our model significantly outperforms the previous approach in both amount-level accuracy and digit-level accuracy.

Table 4: Comparison of the CNN-BiLSTM-CTC model for courtesy amount recognition with previous work.

	No errors	1–error	2–errors	More than 2 errors	Accuracy (digit-level)
Hybrid Rule-Based and Machine Learning System Ahmad (2023)	422 (67%)	125	38	41	87.15%
CNN-BiLSTM-CTC (Ours)	**530 (85%)**	**78**	**12**	**6**	**95.4%**

7. Legal Amount Recognition

The legal amount on Arabic checks denotes the written value and is vital for accurate verification. Its recognition is challenging due to diverse handwriting and the cursive nature of Arabic script. Unlike general handwriting recognition, it is domain-specific, involving numerical expressions and terms such as "Riyal" and "Hallalah." This section outlines the proposed model, including preprocessing, tokenization, and performance evaluation.

7.1. Dataset

The extracted legal amount images from the first phase were used to train the recognition model. Since these images lacked labels, we utilized the corresponding annotations from the CENPARMI dataset, which contains 2,498 manually processed legal amount images—1,779 for training and 719 for testing. Each image is labeled with one or more of 101 Arabic subword classes, representing connected character segments within words. As shown in Fig. 10, the dataset is imbalanced: common subwords such as "فقط" and "لا" appear frequently due to their presence in the phrase "فقط لا غير" ("only the issued amount"), while others like "تسعمئة" and "ستمئة" occur less often. This imbalance results from the multiple valid linguistic forms used to express numeric values in Arabic.

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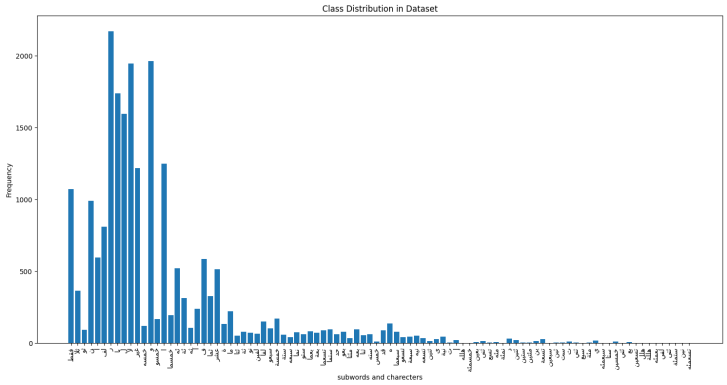

Figure 10: Distribution of labels in the legal amount training dataset, illustrating the frequency of each word, subword and character label.

7.2. A Hybrid CNN-BiLSTM-CTC Model for Legal Amount Recognition

The CNN-BiLSTM model has demonstrated its efficiency in Arabic handwritten text recognition across various datasets and tasks, as evidenced by its performance on the KHATT dataset Lamtougui et al. (2023) and in Mutawa et al. (2024). Therefore, in this work, we employ the CNN-BiLSTM-CTC model for the recognition of Arabic legal amounts.

Similar to the CNN-BiLSTM-CTC model in the courtesy amount recognition, we used a 15 layers CNN to extract the deep features of the images. The output shape of the CNN is (84,512) which represent the main extracted features from the input images. The third step involves feeding these features into two layers of bidirectional long short-term memory (BiLSTM), which are capable of capturing both forward and backward relationships in the sequential input.

In the subsequent step, the output from the final BiLSTM layer is input into a fully connected layer that utilizes a softmax activation function. The output of this layer is a sequence that is 84 characters long, and each element in the sequence corresponds to one of the 24 potential classes that describe the characters and space. On the top of fully connected layer we added a CTC layer to handle the alignment between the input sequence and the predicted output which computes all possible alignments by generating all possible paths to the target sequence and then calculates the probability over all possible paths using dynamic programming. In Fig. 11 we demonstrate the whole structure of the proposed model for legal amount recognition.

The CNN-BiLSTM-CTC model was trained for 500 epochs with early

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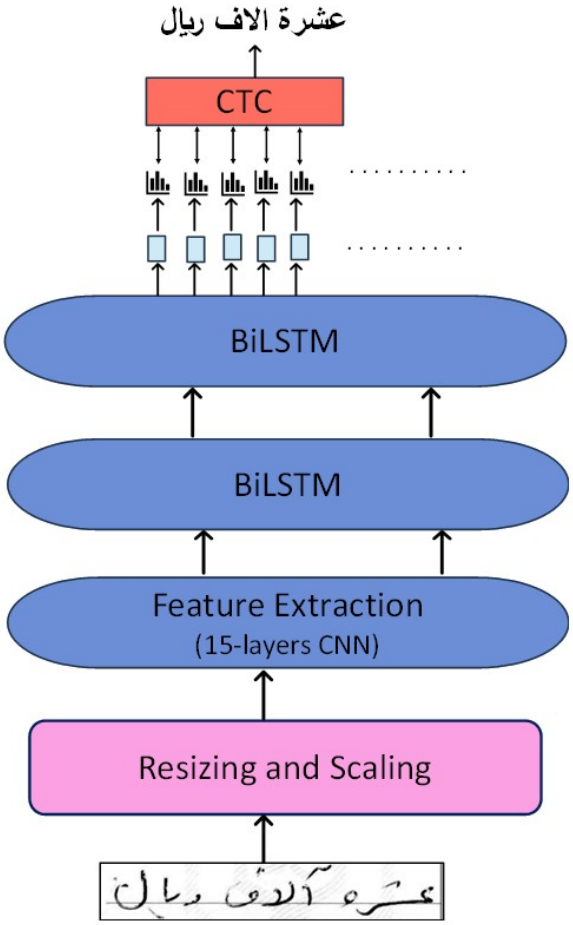

Figure 11: Proposed model for legal amount recognition using CNN, BiLSTM and CTC.

stopping based on validation loss, halting training after 20 epochs without improvement to prevent overfitting. A learning rate decay schedule was applied, starting at 1×10^{-4} and reduced by a factor of 0.9 every 10,000 steps, allowing rapid initial learning followed by finer updates near convergence. The legal amount dataset comprised 1,776 training and 719 testing images, with the training set split into 85% for training (1,335 images) and 15% for validation (267 images). Label sequences were standardized to a uniform length of 84 characters using padding, given a maximum original length of 71.

7.3. Legal Images Preprocessing

To enhance the training process, several preprocessing steps were applied to standardize the input images and labels. All images were resized to (1340, 140) to ensure uniform dimensions, and pixel values were normalized to the range [-1, 1] to facilitate smoother convergence during training. Since Arabic is written right to left and the labels follow this orientation, images were also flipped horizontally to align correctly with the target label sequences. These steps significantly improved data consistency and supported more effective learning of the image-to-sequence mapping.

7.4. Tokenization

The tokenization process in this study involves segmenting the corresponding labels into units referred to as tokens. In the CENPARMI dataset, the labels are based on subword tokenization, where each sequence of connected characters is treated as a single subword. For example, the word "ريال" is assigned three labels: the isolated character "ر" as the first label, the connected characters "يا" as the second label, and the letter "ل" as the third label.

However, Arabic has a complex writing system with distinct rules for character connectivity. Some Arabic characters are written in a connected form, while others remain isolated. Additionally, the shape of a character changes depending on its position within a word. For instance, the letter "ي" appears differently in the word "ريال" compared to its appearance in "خمسين"; in the first case, it is connected to one letter on the left, whereas in the second, it is connected to letters on both sides. As a result, labeling Arabic words based on subword units helps maintain the consistency of character shapes by accounting for their positional variations within words.

On the other hand, some Arabic characters share nearly identical shapes, with only slight differences that do not alter word meaning or structure but rather affect pronunciation and grammar. For example, the letters "ة" and "ه" in the word "ثلاثة" are sometimes written interchangeably, leading to variations in labeling. In the CENPARMI dataset, this results in two different labels, "ثة" and "ثه", both of which represent the same part of the word without affecting its numerical value. Since this study focuses on recognizing the numerical values of written amounts rather than minor character variations, we merged labels that did not impact the final value. Consequently, the number of labels was reduced from 101 to 89.

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Additionally, a second tokenization approach was explored using character-based tokenization. While Arabic character shapes vary depending on their position in a word, potentially confusing the model when extracting representative features, character-level tokenization increases the number of samples per character class, hence, character-based tokenization results in only 24 labels (23 Arabic letters plus a space) compared to the 101 subword tokens.

On the other hand, due to the limited vocabulary, we explored a third tokenization technique: word tokenization. In this method, each class represents a complete written number. The vocabulary encompasses all numerical values (ones, tens, hundreds, and thousands), as well as Saudi currency units such as "ريال" and "هللة", resulting in 89 distinct words. Furthermore, to enhance performance, we merged less frequent classes with those representing the same value but containing minor differences. For example, the words "خمسمئة" and "خمسمائه" both represent the same value but are written differently, with one having fewer samples. By merging these, we reduced the total number of labels from 89 to 45.

To summarize, the following tokenization procedures were experimented in this work:

- Subword tokenization: The original tokenization method used in the CENPARMI dataset, where connected character sequences form subword units.
- Enhanced sub-word tokenization: Sub-words that share the same shape and differ only in diacritical marks or surrounding dots are merged into a single class.
- Word-level tokenization: Subwords were merged into complete words corresponding to the written amounts in the dataset.
- Enhanced word-level tokenization: Words sharing the same value are merged into a single class.
- Character-level tokenization: word labels were split into individual character labels, reducing the number of labels while increasing the number of samples per class.

Table 5 presents a legal amount label tokenized using five different methods, representing the various tokenization approaches explored in this paper. The results of these experiments will be discussed in detail in the results section.

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Table 5: Different tokenization techniques applied to a single legal amount label.

Tokenization	**Label**
Subword tokenization	[أ # لف # و # خمسمائة # و # عشر # و # ن # ر # يا # ل]
Subword tokenization (removing surrounding special characters)	[ا # لف # و # خمسمائه # و # عشر # و # ن # ر # يا # ل]
Word-based tokenization	[الف # و # خمسمائه # و # عشرون # ريال]
Enhanced word-based tokenization	[الف # و # خمسمئه # و # عشرون # ريال]
Character tokenization	[ا # ل # ف # " " # و # " " # خ # م # س # م # ئ # ه # " " # و # " " # ع # ش # ر # و # ن # " " # ر # ي # ا # ل]

7.5. Results and Analysis

In this section, we present a comprehensive analysis of the model's performance using different tokenization techniques. We then discuss the impact of hyperparameter fine-tuning across various configurations. Additionally, we conduct an ablation study to evaluate the model's performance after applying data augmentation in addition to a comparison with applying CNN-BiLSTM model on both manually extracted and automatically extracted legal amounts. Finally, we conduct a pilot experiments using LVLM on the same dataset.

7.5.1. Performance Metrics

To evaluate the model's performance, we employ the Levenshtein distance algorithm, the same method used for assessing the courtesy amount recognition model. This algorithm determines the minimum number of edits—insertions, deletions, and substitutions—required to transform the predicted text into the ground truth.

The model's performance is evaluated at two levels:

- **Character Error Rate (CER):** compares the predicted legal amount to the true legal amount at the character level. It does so by counting the total number of insertions, deletions, and substitutions (as determined by the Levenshtein distance) and dividing this by the total number of characters in the true text. The formula for CER is given in Equation 4:

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$$\text{CER} = \frac{S + D + I}{N} \tag{4}$$

where:

- S = Number of substituted characters
- D = Number of deleted characters
- I = Number of inserted characters
- N = Total number of characters in the ground truth.

- **Word Error Rate (WER):** evaluates accuracy by comparing the predicted value to the true value at the word level. The formula for WER is shown in Equation 5:

$$\text{WER} = \frac{S + D + I}{N} \tag{5}$$

where:

- S = Number of substituted words
- D = Number of deleted words
- I = Number of inserted words
- N = Total number of words in the ground truth.

7.5.2. Final Results Analysis

After enhancing the model performance by experimenting different tokenization techniques and different hyperparameters achieving the optimal performance, we applied the CNN-BiLSTM-CTC model on the test set for legal amount which consists of 719 extracted samples. In table 6 we present the model results on the testing dataset.

While the results indicate a high number of samples with more than two errors on character-level, the number is lower when evaluated at the word level. This is expected, as multiple character errors can occur within the same word, leading to an increased character error count. However, our primary concern is word-level prediction, as the words will later be converted into numerical representations for the verification process. Fig. 12 indicates the confusion matrix for Arabic character-level predictions reveals a notable

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Table 6: Summary of the final results on the test set for legal amount recognition.

Metric	**Value**
CER	5.9%
WER	12%
Number of Correct Samples	400 (55.6%)

presence of insertion and deletion errors, along with some substitution errors at the character level. However, our primary concern lies in word-level substitution errors—instances where an entire word is misclassified as a completely different word. Fortunately, such errors are relatively rare. This observation is particularly important, as it provides a foundation for understanding the effectiveness of our enhancement algorithm discussed in the subsequent section.

At the word level, 400 samples were predicted completely correctly, representing approximately 55% of the testing dataset. Additionally, 161 samples contained a single word error. Fig. 13 presents examples of samples with only one word error. Some samples contain errors in words such as لا and ريال, as seen in the legal amounts one and four. However, these words do not represent numerical values. Therefore, for the purpose of conversion, errors in such words do not affect the final results.

On the other hand, in some cases—such as samples 2 and 3—mispredicting certain characters can result in words that are visually similar in shape but have different numerical values, potentially impacting the final conversion. For example, the word ستمائه (600) was mispredicted as تسعمائه (900). Although the model predicts character by character, the cursive nature of Arabic writing and the lack of clarity in Arabic handwriting can sometimes lead to confusion between characters, resulting in different words.

On the other hand, 70 samples from the dataset contained predictions with two word errors. Fig. 14 presents examples of these samples.

In some cases, we observe missing words such as لا, و, and ريال, as well as missing characters in words like غير. Since these words do not carry numerical value, their omission does not impact the final conversion. However, in sample 2, we observe an incorrect insertion of a space in the number ثلاثون (30), causing it to be split into two sub-words: ثلا and ثون, neither of which correspond to a valid numerical value.

Additionally, in some samples, we noticed insertions or deletions of single

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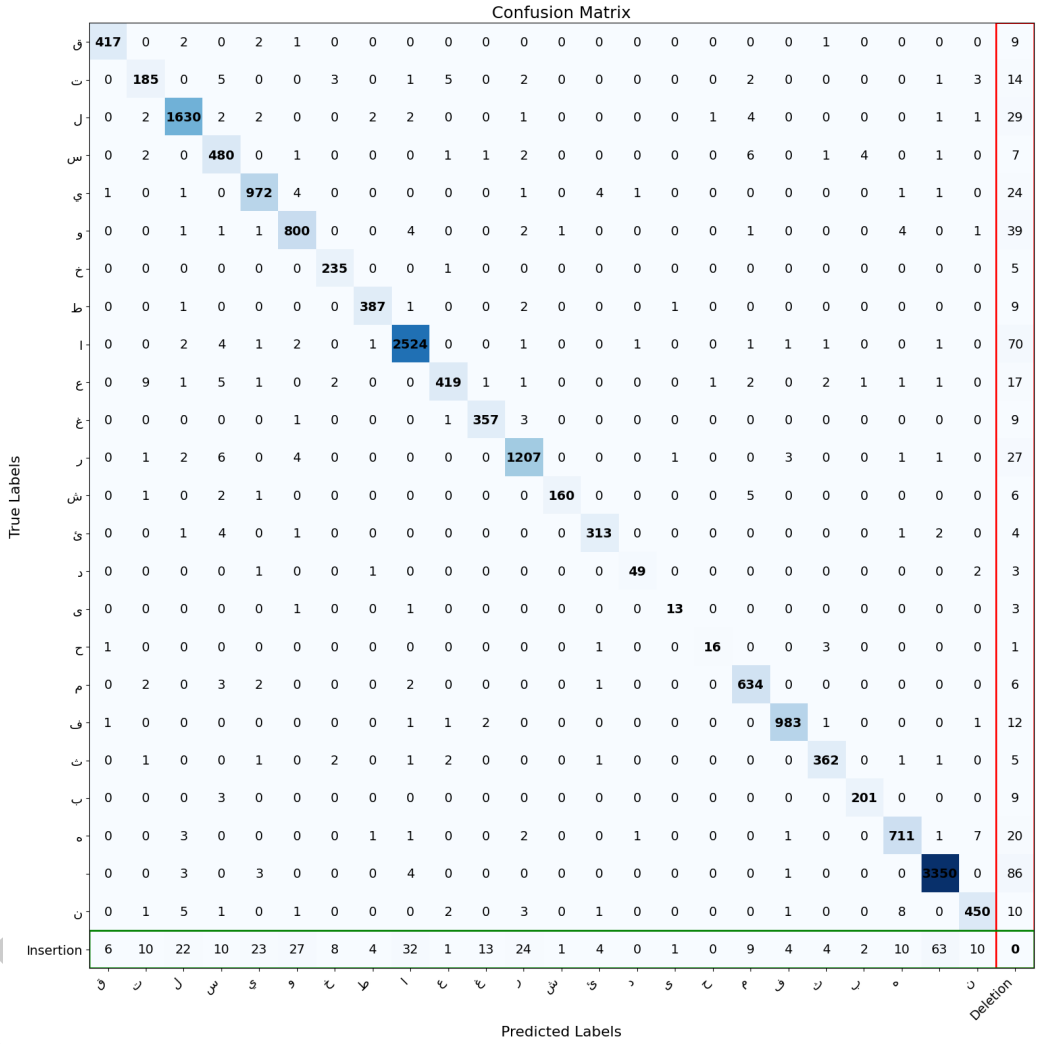


Figure 12: Confusion matrix for character recognition using the BiLSTM-CTC model, including insertions and deletions.

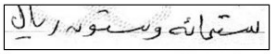
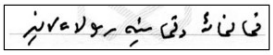
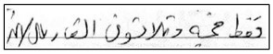
Sample	True Label	Predicted Label
	الفان و مائه و ستون ريال	الفان و مائه و ستون
	ستمائه و ستون ريال	تسعمائه و ستون ريال
	ثمانمائة و ثمانية ريالا لا غير	ثمانمائة و ثمانمئة ريالا لا غير
	فقط خمسة و ثلاثون الف ريال لا غير	فقط خمسة و ثلاثون الف ريال ل غير
	ستمائه ريال لا غير	مائه ريال لا غير

Figure 13: Samples containing a single word-level error, where red indicates a deleted, inserted, or substituted word.

characters within numerical words, such as the omission of the letter ا in اربعه (four) in sample 3. Solutions to such issues will be discussed in the conversion section. Nevertheless, despite the overlap of some words in the written legal amount, the model demonstrated its ability to predict characters accurately. For instance, in sample 4, the word مائتان is written on top of the word و. However, the model was still able to correctly predict both characters.

On the other hand, 86 samples contained more than two errors, exceeding the number of samples with exactly two errors. Fig. 15 presents examples of these cases. Some samples were not extracted correctly, resulting in incomplete amounts and missing words, as seen in sample two.

Moreover, certain legal amounts included handwritten text that was not labeled because it is not part of the predefined dictionary. For instance, in sample one, the end of the check contains the phrase دفعه على الحساب, which indicates a payment reason rather than a numerical representation of the amount. This led to multiple errors in the prediction. In addition to some errors in inserting or deleting charecters from words which lead to word error.

The overall word-level accuracy of the model was 88% on word level and 94.1% on charecter level. However, many of the errors occurred in words that

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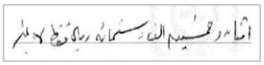
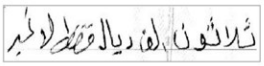
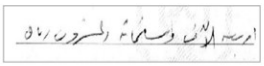
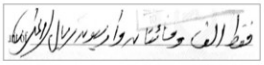
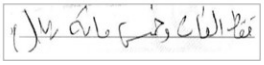
Sample	True Label	Predicted Label
	اثنان و خمسين الف و ستمائه ريال فقط لا غير	اثنان و خمسون الف ستمائة ريال فقط لا غير
	ثلاثون الف ريال فقط لا غير	ثل اثون الف ريال فقط لا غير
	اربعه الاف و ستمائه و عشرون ريال	ربعه الاف و ستمائه و عشرون
	فقط الف و مائتان و أربعون ريال لا غير	فقط الف و مائتان و اربون ريال غير
	فقط الفان و خمس مائه ريال	فقط الفان و خمسمائه

Figure 14: Samples containing two word-level errors, where red indicates a deleted, inserted, or substituted word.

do not represent numerical values, meaning these mistakes do not impact the verification process and are not counted as critical errors. Nevertheless, the final accuracy of the extracted legal amount depends on the accuracy of the conversion, which will be discussed in the conversion section.

7.5.3. Ablation Studies

This section presents a comparison of model performance both before and after applying data augmentation. Additionally, we evaluate the effectiveness of the YOLO model in extracting legal amount images from check images by comparing the performance of the CNN-BiLSTM model when applied to manually extracted versus automatically extracted legal amount images (using the YOLO-based method).

Due to the limited size of our dataset and the scarcity of publicly available real bank check images, we applied data augmentation to expand the training set. Specifically, we utilized four transformation techniques on each image, effectively increasing the dataset size by a factor of four, resulting in a total of 6,048 images.

Since legal amount images frequently contain lengthy written sentences, we carefully selected augmentation techniques that maintain the integrity of the text while preventing any part of it from being cropped or removed. The applied transformations include zoom-in, left rotation, right rotation,

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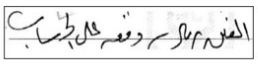
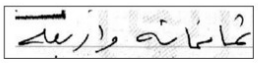
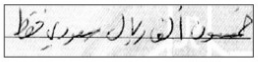
Sample	True Label	Predicted Label
	الفين ريال	الفين و ما ودلى لا غير
	ثمانمائه و اربعه و خمسون	ثمان مائه و غير
	خمسمائه ريال لا غير	خمسمائه و ل ل ريال لا غير
	خمسون الف ريال سعودي فقط	خمسون الف ريال سع عدى ل غير

Figure 15: Samples containing more than two word-level errors, where red indicates a deleted, inserted, or substituted word.

and elastic transformation. We excluded zoom-out, as it was found to crop parts of the written values in some samples. Fig. 16 illustrates the four transformations applied to a single sample using these data augmentation techniques.

Nevertheless, the model's performance significantly improved after applying data augmentation, as shown in Table 7. The number of completely correct predictions increased from 139 to 164 samples, while the Character Error Rate (CER) and Word Error Rate (WER) notably decreased to 4% and 9.3%, respectively. These results demonstrate the effectiveness of data augmentation in enhancing model performance and managing data scarcity issues.

Table 7: Model performance on the validation dataset before and after applying data augmentation, highlighting its impact on overall performance.

Metric	**Before augmentation**	**After augmentation**
CER	5.5%	4%
WER	12%	9.3%
Number of Correct Samples	139(52%)	164(61%)

On the other hand, we evaluated the performance of the CNN-BiLSTM model on automatically extracted images, which were manually extracted

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Figure 16: Four transformations of a single sample generated using four data augmentation techniques: zoom in, right rotation, left rotation and elastic transformation.

and processed in the CENPARMI dataset. Table 8 presents a comparison of the CNN-BiLSTM model's performance on manually extracted legal amount images (from CENPARMI) versus automatically extracted images using a fine-tuned, pretrained YOLO model. The results clearly demonstrate the performance difference between using manually extracted legal amount images and automatically extracted images. While the manually extracted images in the CENPARMI dataset are more processed and cleaned of surrounding noise, the model performs better on automatically extracted images, which contain noise around the written amounts. To provide a clearer understanding, Fig. 17 presents sample cases where the model correctly predicted the amounts in automatically extracted images but misrecognized certain characters in manually extracted images.

Table 8: Comparison of model performance using manually extracted legal amounts versus automatically extracted amounts from a fine-tuned YOLO model.

Metric	**Manually Extracted**	**YOLO Extracted**
CER	10%	5.9%
WER	20%	12%
Number of Correct Samples	274(38%)	400(55.6%)

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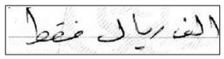
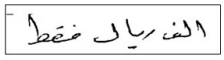
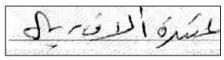
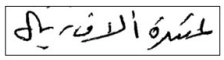
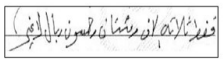
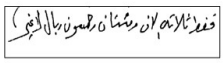
	Sample	Predicted amount
Automatically extracted		الف ريال فقط
Manually extracted		الف ط ريال قط
Automatically extracted		عشره الاف ريال
Manually extracted		عشره الاف ريالسعود
Automatically extracted		فقط ثلاثه الاف و مئتان و خمسون ريال لا غير
Manually extracted		فقط ثلاثه الاف و ثمئتان و خمسون ريال لا غير

Figure 17: Comparison of CNN-BiLSTM-CTC model performance for legal recognition, using images extracted by YOLO versus manually extracted images from the CNEPARMI dataset.

7.5.4. Comparison with Previous Work

Since no prior research targets Arabic legal amount recognition on the CENPARMI dataset, we qualitatively compare our results with studies in other languages (Table 9). HMM-based methods achieved 82.96% for Brazilian and 80.88% for Persian legal amounts, while our Arabic model reached 88%, outperforming the Persian counterpart despite limited data. For Chinese, complex characters led to 93.5% character-level but only 60% amount-level accuracy. Hindi-based approaches achieved 79.9% word-level accuracy, and English systems, aided by large datasets and simpler scripts, reached nearly 99% at the character level. French studies reported 87.1% word-level and 63% amount-level accuracy. In contrast, our end-to-end CNN-BiLSTM-CTC model, operating without segmentation, achieved 88% word-level and 94.1% character-level accuracy, demonstrating robust performance under real-world handwriting and noise challenges.

8. Postprocessing

This section describes the postprocessing of extracted fields, including converting legal amounts from Arabic words to numbers, applying enhancement techniques to improve accuracy, refining courtesy amount predictions

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Table 9: Previous work on legal amount recognition, including studies on Indian checks and sub-word/character recognition.

Previous Work	**Language**	**Model**	**Results**
Singh et al. (2024)	English	CNN	99% (charecter-level)
Jayadevan et al. (2011)	Hindi	GSC-BVM + VPP-DTW hybrid	79.90%(word-level)
de Almendra Freitas et al. (2000)	Brazilian	Hidden Markov Models	82.96%(word-level)
Shahriarpour and Sadri (2014)	Persian	Hidden Markov Models	80.88%(word-level)
Tang et al. (2004)	Chinese	Neural Network/ HMM	93.50%(charecter-level)
Knerr et al. (1997)	French	Log Linear + Neural Network	87.10%(word-level)
Our work	**Arabic**	**CNN-BiLSTM-CTC**	**94.9%(character-level) 88% (word-level)**

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using the converted legal values, and evaluating the overall effectiveness of these steps in amount extraction and recognition.

8.1. Legal Amount Conversion

Conversion involves transforming recognized written amounts into numerical values, a task complicated by Arabic's linguistic rules. For example, 25 is written as "خمسة وعشرون" ("five twenty"), with the ones digit preceding the tens, and numbers like two thousand can appear in multiple forms ("ألفان", "ألفا", "ألفين", "ألفي"). Despite these variations, a limited dictionary covering Arabic numbers and currencies (e.g., riyal, hallala) allowed the use of a rule-based model to convert legal amounts for verification and comparison with courtesy amounts.

In this work, the conversion process was carried out in three main stages. First, a direct match was attempted using a predefined dictionary. Next, minimum edit distance was employed to improve matching accuracy. Finally, the predicted courtesy amounts were utilized to enhance the conversion of legal amounts, and vice versa. On the other hand, the conversion performance was evaluated on the testing dataset, which consists of 719 samples, by comparing the converted values with the true courtesy amount values. Accuracy was then calculated as the ratio of correctly converted samples to the total number of samples. In the following sections we will provide thorough explanations of the three stages of conversion to achieve the optimal performance.

8.1.1. Direct Matching from Dictionary

As the first step in the conversion process, we defined a dictionary containing all possible variations of written numbers across different digit ranges, from ones to millions.

In the proposed conversion algorithm, we assume that a written amount can include three main numerical units:

- Thousands Unit: This unit includes the word "ألف" or any of its variations. Any words appearing before this term are considered part of the thousand multiplier. For example, in the written amount "خمس وعشرين ألفاً", the words preceding "ألفاً"—namely, "خمس وعشرين"—are part of the thousand unit.

- Hundreds Unit: When numbers involve the word "مئة" or "مائة", the only scenario in which a preceding number is considered part of the same

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unit is when hundred number like "خمسمئة" is written as two separate words, such as "خمس مئة". To handle this case, the algorithm examines the preceding token. If it represents a single-digit number, the hundreds unit is multiplied by this value and added to the total. Otherwise, it is multiplied by one and added to the total.

- Tens and Ones Unit: This unit includes both tens and ones, which are simply added together to form the numerical representation. These units may appear before the hundreds or thousands units, as seen in the thousands unit such as "خمس وعشرون ألفاً" (twenty-five thousand), where "twenty-five" belongs to the thousands unit. Alternatively, they may appear in isolation at the end of the written amount.

In the first proposed algorithm, we iterate through each word of the recognized legal amount to check its availability in the dictionary. If a word is not found in the dictionary, the algorithm assumed its value to be zero. Additionally, non-numerical phrases such as "فقط لا غير" are also assigned a value of zero.

Next, the algorithm determines whether the word represents a specific multiplier, such as thousands or hundreds, in which case the preceding amount is multiplied by this value. Otherwise, if the word does not indicate a multiplier, it is directly added to the total amount to compute the final numerical value.

Algorithm 1 presents the proposed approach for converting legal amounts.

To evaluate conversion performance, we compared the converted values with the corresponding true values of the courtesy amounts. The evaluation was conducted on a testing set consisting of 719 samples, of which 476 were correctly converted, resulting in a conversion accuracy of 65%. However, some minor errors in words led to the omission of entire values from the conversion. For example, a legal amount was predicted as 'سبعه', 'الاف', 'و', 'تسعمائه', 'و', 'خمسه', 'و', 'سعون' instead of 'سبعة آلاف و تسعمائة و خمسة و ستون'. In this case, the word 'ستون' was incorrectly predicted as 'سعون', and the substitution of a single character led to the omission of the entire word from the conversion, resulting in the number being converted as 7905 instead of 7965. To avoid such cases and improve the conversion process, we applied the minimum edit distance approach, which is explained in the next section.

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Algorithm 1 Convert legal amount to numeric value

```
Input: labels (List of Arabic words representing the amount), word_to_number (Dictionary mapping Arabic words to numbers)
Output: total (Numeric representation of the amount)
total ← 0
current_value ← 0
for each word in labels do
    if word ∈ word_to_number then
        if word = 'الف' or word = 'ألف' or word = 'الاف' then
            current_value ← (if current_value ≠ 0 then current_value else 1) × 1000
            total ← total + current_value
            current_value ← 0
        else if word = 'مئة' or word = 'مائه' or word = 'مئه' or word = 'مائة' then
            current_value ← (if current_value ≠ 0 then current_value else 1) × 100
            total ← total + current_value
            current_value ← 0
        else
            current_value ← current_value + word_to_number[word]
        end if
    end if
end for
total ← total + current_value
return total
```

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8.1.2. Matching from Dictionary Using Minimum Edit Distance

Since direct matching can result in the omission of an entire word representing a digit or number due to a single character error, we employed the minimum edit distance approach to ensure that all words in the legal amounts are considered.

The updated legal amount conversion algorithm processes the legal amount word by word, searching for each word in the dictionary to retrieve its corresponding value. However, if a word is not found in the dictionary, the algorithm calculates the minimum edit distance between the word and every entry in the dictionary using the Levenshtein distance. This metric determines the minimum number of edits required to transform the predicted word in the legal amount into a dictionary word. In cases where multiple words have the same minimum edit distance, the algorithm selects the first word that appears in the dictionary.

On the other hand, to address the issue of missing spaces between words, which can result in the merging of two separate words, we implemented a linear search approach. For example, consider the true legal amount "عشره الاف ريال". If the space between عشره and الاف is not predicted correctly, the two words may be merged into "عشرهالاف ريال" When attempting to match عشرهالاف with dictionary words, this could lead to an incorrect prediction, such as identifying it as عشره instead.

To mitigate this issue, we applied a linear search mechanism. If a word is not found in the dictionary, the algorithm systematically checks for possible subword matches by progressively segmenting the word. It begins by checking the first character against the dictionary. If match is found, it combines the first and second characters and checks again. If still matched, it continues by adding the third character, then the fourth, and so on, until a not valid dictionary match is identified and the word will be split on this place. This linear search method can effectively addresses errors caused by incorrect space predictions, improving the accuracy of the conversion process.

Using minimum edit distance and linear search to select words from dictionary has improved the conversion performance. As the number of correctly converted legal amounts has increased from 476 to 557 sample which mean increasing the accuracy of legal amount recognition from 66% to 77%. This outcome is expected, as the enhanced method considers all written words in the legal amount.

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8.2. Enhancing Legal Amounts Conversion Using Courtesy Amounts

Using the minimum edit distance approach resolved the issue of omitting an entire word due to one or more character errors. However, when applying this method, the algorithm selects words from the dictionary that have the smallest edit distance from the predicted word. In some cases, the algorithm returns a list of multiple words with the same minimum edit distance, which may include one, two, or more options. Previously, the selection process was based on choosing the first word from the returned list.

To improve this approach, we enhanced the algorithm to make a more informed selection by incorporating the predicted courtesy amount. Instead of automatically selecting the first option, the algorithm now evaluates all possible words in the returned list and compares them with the predicted courtesy amount. If any of the options match the predicted courtesy amount, that word is selected. If none of the options match, the algorithm defaults to selecting the first word in the list. This refinement improves the accuracy of the legal amount conversion by leveraging additional information from the courtesy amount. In Table 10 we present some samples conversion and the process of converting them using predicted courtesy amount.

Table 10: Legal amount conversion using minimum edit distance and predicted courtesy amount.

Predicted Legal Amount	['ال', 'ياد', 'فقط', 'لا', 'غير']
True Legal Amount	['الف', 'ريال', 'فقط', 'لا', 'غير']
Predicted Courtesy Amount	1000
Candidates Labels and their conversion	('الف', 'واحد', 'فقط', 'لا', 'غير'), 1001) ('الف', 'احد', 'فقط', 'لا', 'غير'), 1001) ('الف', 'ريال', 'فقط', 'لا', 'غير'), 1000) ('الف', 'لا', 'فقط', 'لا', 'غير'), 1000)
Selected Candidate	1000

The enhanced algorithm using predicted courtesy amounts has improved the recognition results to achieve 601 correctly converted sample out of 719 which is 83% accuracy.

8.3. Enhancing Courtesy Amounts Recognition Using Legal Amounts

In the Indian numeral system, the zero digit is represented as a small period, and a missing zero can drastically change a value, e.g., from thousands to hundreds. To improve courtesy amount predictions, we leveraged

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converted legal amounts as a reference, similar to how a bank employee verifies unclear digits. Our algorithm first compares the lengths of predicted legal and courtesy amounts, then calculates the minimum edit distance. If the distance is one and caused by an insertion or deletion of digit zero, the courtesy amount is adjusted accordingly. We specifically focus on zero-digit insertion and deletion errors, as legal amount predictions are generally more reliable for determining the number of digits, particularly the unit of the amount. Errors in recognizing the amount unit in the legal amount typically require the misrecognition of an entire word, which involves multiple characters. For example, as demonstrated in Table 11, a predicted courtesy amount may miss a zero or insert a zero digit, significantly altering the value. However, for a similar error to occur in the legal amount, the model would need to misrecognize the entire charecters of word آلاف (thousands) and replace it with charecters of word مئة (hundred). This highlights why the legal amount prediction is more reliable when determining the correct number of digits in the amount. On the other hand, we did not rely on the legal amounts for substituting any predicted digits as in some cases legal amount prediction can replace digits due to misrecognizing charecters of written number so predicted legal amounts can be reference for deciding the predicted amount.

Table 11: Enhanced courtesy values after using predicted and enhanced legal amounts.

	Example 1	**Example 2**	**Example 3**
True Courtesy	10000	2000	4000
Predicted Courtesy	1000	200	40000
Predicted Legal	عشره الاف ريال فقط لا غير	فقط الفين ريال لا غير	اربع الاف ريال لا غير
converted Legal	10000	2000	4000
Transform Operation	(INSERT, 0)	(INSERT, 0)	(DELETE, 0)
New Courtesy	10000	2000	4000

this method has enhance the predicted courtesy amounts and increase the number of correctly predicted courtesy amounts to 622 instead of 613 out of 719 which demonstrates the effectiveness of using legal amounts for enhancing courtesy prediction by inserting and deleting missing digits.

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8.4. Final Results

This section presents the final evaluation results of the proposed system for recognizing legal and courtesy amounts from Arabic bank checks. The evaluation focuses on the effectiveness of the postprocessing phase, which includes conversion of subword predictions into numerical values and enhancement techniques aimed at improving recognition accuracy. Table 12 summarizes the core performance metrics, including character and word error rates, recognition accuracy at both digit and amount levels, and the overall rate of correctly recognized check amounts.

Table 12: Summary of final recognition results after postprocessing

Courtesy Amount Recognition	
Digit-level accuracy	96.1%
Amount-level accuracy	86.5%
Legal Amount Recognition	
Character Error Rate (CER)	5.4%
Word Error Rate (WER)	12%
Legal Amount Recognition Accuracy	83%
Final results	
Legal-Courtesy Matching Accuracy	78.5%

The system achieved 96% digit-level accuracy for courtesy amounts, demonstrating robustness against handwriting variability and noise, and 86.5% accuracy at the full-amount level after incorporating legal amounts for enhancement. For legal amounts, the complexity of handwritten Arabic led to a 5.4% Character Error Rate and 12% Word Error Rate; however, postprocessing and contextual verification mitigated these errors. Overall, legal-to-courtesy conversion accuracy reached 83%, showing the system's ability to extract coherent and semantically correct numerical information from written amounts.

These results highlight the importance of postprocessing in improving prediction quality and ensuring reliable amount extraction, which is essential for verification — the primary goal of recognizing legal and courtesy amounts. To evaluate system courtesy and legal amount verification, enhanced courtesy amounts were compared to converted legal amounts and, when matched, to the true values. Out of 719 Arabic check samples, 563 were correctly verified, resulting in a verification accuracy of 78.5%.

To further evaluate the system's reliability and to analyze the consistency of the verification process, we examined the occurrence of false positives.

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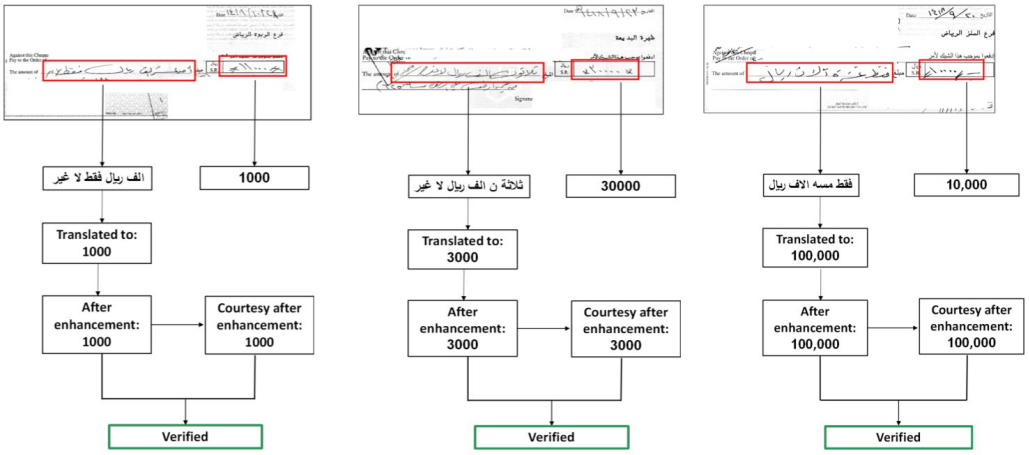

Figure 18: Samples of false positive cases where the model predicted matching legal and courtesy amounts that do not correspond to the true values.

In a real world scenario, a false positive refers to a case where the system incorrectly indicates that the legal and courtesy amounts match, despite they are not matched on the check. In our dataset, however, all original check images contain legal and courtesy amounts that are truly equal and verified. Therefore, in our context, a false positive is defined as a case in which the model predicts matching legal and courtesy amounts, but these predicted values do not correspond to the actual ground truth values.

Among the 154 samples that were not successfully verified, 10 instances were identified as false positives according to the previous definition of false positive. In these cases, the system produced a match between the legal and courtesy amounts, but the matched value was incorrect. Examples of such misclassified checks are illustrated in Fig. 18. These cases highlight scenarios where the system output appears correct from a verification standpoint, yet fails in terms of matching to the ground truth amount.

8.4.1. Comparison with previous work

To the best of our knowledge, no previous work has developed a full pipeline for extracting and recognizing both legal and courtesy amounts across languages. We compared our Arabic check recognition system with two commercial solutions—Parascript CheckUltra Parascript (2022) and A2iA CheckReader Gorski et al. (2001). While these systems achieve high accu-

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racies (over 98% STP for Parascript and 65–85% for A2iA) in Latin-based scripts, they lack native Arabic support. Our proposed model, tailored for Arabic checks, achieved 96% accuracy on handwritten digits, 88% on words, and 78.5% full amount recognition with only 1.4% false positives. Despite slightly lower verification accuracy, it demonstrates strong real-world robustness and addresses the unique challenges of Arabic script recognition.

9. Conclusion and Future Work

This paper presents an end-to-end system for automating Arabic bank check processing by extracting, recognizing, and verifying legal and courtesy amounts. Using a YOLO-based model, we accurately localized both amount fields, achieving an accuracy of 100% for courtesy amounts and 99.6% for legal amounts under a 50% IoU threshold. The recognition stage employed a CNN-BiLSTM model, which reached 87% accuracy for courtesy amounts and 88% at the word level for legal amounts using character-based tokenization.

To bridge the semantic gap between the two amount types, we developed a legal amount conversion algorithm that iteratively refined its output by leveraging the predicted courtesy amounts. This mutual enhancement helped verify the consistency between the amounts, resulting in an overall check verification accuracy of 78.5%. This is the first known end-to-end system tested on the CENPARMI Arabic check dataset, demonstrating practical application of deep learning in financial document processing for Arabic-script text.

While the system shows promising results, it is limited by the size and diversity of the dataset used. Training on a single, relatively small dataset limits the generalizability of the models to new handwriting styles, check layouts, and image qualities. Additionally, the system's performance is dependent on successful early-stage extraction, making it vulnerable to cascading errors.

Future work can focus on incorporating transformer-based models and pretrained architectures like ResNet or VGG to enhance recognition accuracy. Exploring large vision-language models (LVLMs) fine-tuned for Arabic handwriting also holds potential. Expanding the dataset and improving model robustness will be essential for deploying this system in real-world banking environments.

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